Lesson notes · for you

Variables and input

Programming · Level 2 · Lesson 2

Time	Materials
20 min	A phone with this lesson downloaded; A notebook

Learning goals

- Store and change values in variables.
- Ask with input() and turn text into a number with int().
- Explain why "5" + "3" is "53".

Activity: Label the boxes

1. Warm-up (3 min): Draw a box on paper, label it score and write 20 inside. That is score = 20.
2. Trace (7 min): Follow score = 20, score = score + 5, score = score * 2. After each line, cross out the old value and write the new one.
3. Predict (7 min): Write what each prints: print("5" + "3"), print(5 + 3), print(int("5") + 3).
4. Check (3 min): Run them on the phone and compare with your predictions.

Paper boxes work well: a variable is a box that holds one value at a time.

Discussion

- Why does input() give text, not a number?
- What does score = score + 5 mean in words?
- Which error have you met today?

Check yourself

- Write two lines: ask for a number, and print it doubled.

Try it this week

Ask a friend to predict: x = 5, then x = x * 2, then print(x). Then run it. Who was right?

Worksheet

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Name _____ Class _____ Date _____

1 **You type 13. What happens?**

```
age = input("Age? ")
print(age + 1)
```

◯ It prints 131 ◯ An error: text plus a number ◯ It prints 14

2 **Finish the program: it asks for a name and an age, then prints a sentence like Amina will be 14 next year.**

```
name = input("Name? ")
# ask for the age, then print the sentence
```

Write your program:

3 **This program stops with a TypeError. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  name = input("Name? ")
2  goals = 3
3  print(name + " scored " + goals + " goals")
```

4 **Fill the gaps: the program starts with 10 coins, adds 5, and prints 15.**

```
coins = 10
coins = coins _____ 5
print( _____ )
```

Word bank: `-` `+` `"coins"` `coins`

5 **What does this program print?**

```
x = "5"
y = "3"
print(x + y)
```

◯ 53 ◯ An error ◯ 8

✎ **Which question would you like a program to ask you, and what should it answer?**

Check your answers

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1 **An error: text plus a number**

age is the text "13", and text + a number is a TypeError.

2

```
name = input("Name? ")
age = int(input("Age? "))
print(name, "will be", age + 1, "next year")
```

input() gives text, so the age goes through int() before age + 1.

3 **Line 3: `print(name, "scored", goals, "goals")`**

+ can join text to text, but not text to a number: goals is 3, a number.

4

```
coins = 10
coins = coins + 5
print(coins)
```

coins + 5 adds to the old value and stores 15.

5 **53**

x and y are text (they are in quotes), and + joins text: "5" + "3" is "53".